

# Unit 2: Natural Resources, Energy & Conservation

## Handwritten Student Notebook

### Syllabus Topics Covered:

OFFICIAL SYLLABUS TOPICS COVERED  
Resource Foundations: Renewable vs. non-renewable resources, sustainable management  
Forest Resources: Uses, deforestation causes, mining effects, dams & tribal displacement, Chipko/Appiko  
Water Resources: Over-exploitation, groundwater...

### Table of Contents

|                                                               |       |
|---------------------------------------------------------------|-------|
| 1. Natural Resources & Sustainability                         | p. 2  |
| 2. Forest Resources & Deforestation                           | p. 2  |
| 3. Water Resources & Rainwater Harvesting                     | p. 3  |
| 4. Mineral Resources & Mining Impacts                         | p. 4  |
| 5. Food Resources & Modern Agriculture                        | p. 4  |
| 6. Land Resources & Soil Erosion                              | p. 5  |
| 7. Energy Resources (Fossil Fuels vs. Renewable Alternatives) | p. 6  |
| 8. Solved Environmental Calculations                          | p. 7  |
| 9. Quick Revision Sheet                                       | p. 10 |
| 10. Authentic Exam Archive & Model Solutions                  | p. 11 |
| 11. Prescribed References                                     | p. 19 |

## 1. Natural Resources & Sustainability

### Definition:

A **Natural Resource** is any raw material, physical matter, or energy flow provided by the Earth that is essential or useful for sustaining human civilization and ecological integrity.

| Classification                                               | Definition & Replenishment Rate                                                                                          | Core Examples                                                                                                           |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Renewable Resources</b><br>(Inexhaustible / Regenerative) | Resources replenished naturally through biogeochemical or physical cycles at a rate equal to or faster than consumption. | Solar radiation, wind kinetic energy, tidal flow, biomass (when harvested below maximum sustainable yield), freshwater. |
| <b>Non-Renewable Resources</b><br>(Exhaustible)              | Finite geological reserves formed over millions of years that cannot regenerate on human timescales once extracted.      | Fossil fuels (coal, petroleum, natural gas), metallic ores (iron, bauxite, copper), uranium, non-metallic minerals.     |

### Quick Recall:

Renewable = regenerated by natural cycles (solar, wind, water). Non-renewable = finite geological reserves (fossil fuels, metallic minerals).

## 2. Forest Resources & Deforestation

Forests are Earth's lungs, covering approximately **31%** of global land area. They serve critical economic functions (timber, pulp, resin, medicinal herbs) and invaluable ecological services (carbon sequestration, oxygen generation, hydrological regulation, soil stabilization).

### 2.1 Deforestation: Causes & Ecological Fallout

- **Principal Causes:** Agricultural expansion (slash-and-burn / shifting cultivation), commercial timber logging, open-cast mining in forested belts, large-scale hydroelectric reservoirs, urban infrastructure corridors.
- **Consequences:** Accelerated soil erosion, catastrophic loss of biodiversity, disruption of regional microclimate and rainfall cycles, increased atmospheric  $CO_2$ , and displacement of indigenous forest communities.

## 2.2 Dams & Forest Conservation Movements

- **Dams & Tribal Displacement:** Mega-dam projects submerge thousands of hectares of prime virgin forest (e.g. Sardar Sarovar, Tehri Dam) and forcibly displace indigenous tribal populations without adequate rehabilitation.
- **The Chipko Movement (1973, Uttarakhand):** Led by Sunderlal Bahuguna and Chandi Prasad Bhatt; villagers (notably rural women) hugged trees to prevent commercial loggers from cutting them down.
- **The Appiko Movement (1983, Western Ghats, Karnataka):** Inspired by Chipko, villagers in Uttara Kannada embraced forest trees with the slogan "*Ubsu, Belasu, Balasu*" (Save, Grow, and Wisely Use).

### Quick Recall:

Deforestation causes soil loss, climate shifts, and tribal displacement. Grassroots conservation: Chipko (Uttarakhand, Sunderlal Bahuguna) and Appiko (Karnataka Western Ghats).

## 3. Water Resources & Rainwater Harvesting

Freshwater constitutes only 2.5% of global water reserves. Human extraction for agriculture (70%), industry (20%), and municipal use (10%) is creating severe global water scarcity.

### 3.1 Over-utilization & Groundwater Depletion

- **Over-pumping:** Deep tube-wells extract groundwater faster than monsoon recharge rates, plunging water tables downward by meters per year in agricultural breadbaskets (Punjab, Karnataka).
- **Land Subsidence & Contamination:** Excessive pumping causes aquifer compaction, coastal saltwater intrusion, and mobilization of toxic geogenic fluoride and arsenic into drinking supplies.
- **Inter-State River Conflicts:** Severe geopolitical disputes over shared river basins during dry spells (e.g. the **Cauvery River Dispute** between Karnataka and Tamil Nadu, and the **Krishna Water Dispute**).

### 3.2 Rainwater Harvesting (RWH)

Rainwater harvesting captures and stores rainwater for direct use or for recharging subterranean aquifers via recharge pits and borewells.

Formula: Annual Rooftop Rainwater Harvest Potential:

$$V = A \times R \times C$$

where:

- $V$ : Total harvestable water volume ( $m^3$ ).
- $A$ : Catchment roof area ( $m^2$ ).
- $R$ : Annual rainfall ( $m$  or  $mm/1000$ ).
- $C$ : Runoff coefficient (fraction of rainfall not absorbed;  $C \approx 0.80 - 0.90$  for concrete roofs,  $0.60$  for tiled roofs).

Quick Recall:

Over-extraction causes aquifer depletion and river disputes (Cauvery). Rooftop harvesting yield formula:  $V = A \times R \times C$  ( $m^3$ ).

## 4. Mineral Resources & Mining Impacts

Definition:

A **Mineral** is a naturally occurring, inorganic solid substance possessing a definite chemical composition and an ordered internal crystalline atomic structure.

### 4.1 Environmental Fallout of Mining

- **Landscape Destruction:** Open-cast surface mining strips overburden vegetation, creating massive craters and mine dumps.
- **Acid Mine Drainage (AMD):** When sulfide minerals (e.g. pyrite,  $FeS_2$ ) are exposed to air and water during mining, bacterial oxidation generates sulfuric acid ( $H_2SO_4$ ), which leaches toxic heavy metals into streams:



- **Air & Noise Pollution:** Blasting, drilling, and hauling generate fugitive particulate matter ( $PM_{2.5}$ ,  $PM_{10}$ ) causing silicosis in miners.
- **Tailings Dam Failure:** Collapse of wet tailing ponds releases millions of tons of toxic slimes into river networks.

Quick Recall:

Minerals are naturally occurring inorganic solids. Mining causes land degradation, heavy metal toxicity, and Acid Mine Drainage (sulfide oxidation generating  $H_2SO_4$ ).

## 5. Food Resources & Modern Agriculture

The Green Revolution dramatically increased grain yields through High-Yield Varieties (HYVs), synthetic fertilizers, chemical pesticides, and intensive canal irrigation, but triggered severe ecological side effects:

| Modern Agricultural Practice                            | Environmental Degradation Mechanism                                                                                                                                                | Ecological / Health Impact                                               |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Excessive Synthetic Fertilizers (Nitrates & Phosphates) | Runoff into surface lakes causes <b>Eutrophication</b> (algal blooms, oxygen depletion, fish kills).<br>Groundwater leaching causes <b>Blue Baby Syndrome</b> (Methemoglobinemia). | Aquatic dead zones; impaired oxygen delivery in infants.                 |
| Persistent Organochlorine Pesticides (DDT)              | Lipid-soluble, non-biodegradable toxins accumulate up food chains via <b>Biomagnification</b> (concentration amplifies $\times 10^5$ from water to apex raptors).                  | Eggshell thinning in birds, endocrine disruption, human carcinogenicity. |
| Intensive Canal Flood Irrigation                        | Excessive water raises water table close to surface ( <b>Waterlogging</b> ). Evaporation concentrates dissolved mineral salts in topsoil ( <b>Salinization</b> ).                  | Osmotic stress, plant root asphyxiation, cropland desertification.       |

### Quick Recall:

Agricultural side effects: Eutrophication & Blue Baby Syndrome (nitrates); Biomagnification (DDT concentration amplification); Waterlogging & Salinization (over-irrigation).

## 6. Land Resources & Soil Erosion

Soil is a non-renewable resource on human timescales, requiring centuries to form a single centimeter of topsoil through bedrock weathering.

### 6.1 Mechanisms & Types of Soil Erosion

A. Wind Erosion (Exam Focus: CIE 1 2024 Question):

- **1. Saltation (50 — 70% of wind erosion):** Fine-to-medium sand particles (0.1 — 0.5 mm) bounce along the surface in short hops, dislodging other particles upon impact.
- **2. Surface Creep (5 — 25% of wind erosion):** Larger particles (0.5 — 2.0 mm) too heavy to be lifted are rolled or pushed along the ground by colliding saltating grains.

- **3. Suspension (3 – 15% of wind erosion):** Extremely fine silt and dust particles ( $< 0.1 \text{ mm}$ ) are lifted high into the atmosphere and transported thousands of kilometers as dust storms.

#### B. Water Erosion:

- **Sheet Erosion:** Uniform detachment and thin horizontal removal of topsoil by overland sheet runoff.
- **Rill Erosion:** Concentrated runoff cuts small, shallow, distinct finger-like channels (rills).
- **Gully Erosion:** Rills converge into deep, steep-sided channels (gullies) that dissect croplands into uncultivable ravines (badland topography).

#### 6.2 Soil Conservation Measures

- **Contour Bunding & Terracing:** Sculpting steep slopes into stepped terraces to slow surface runoff velocity.
- **Afforestation & Shelterbelts:** Planting dense tree windbreaks perpendicular to prevailing winds to block wind erosion.
- **Mulching & Cover Cropping:** Retaining crop residues on soil to prevent direct raindrop impact detachment.

#### Quick Recall:

Wind erosion types: Saltation (bouncing grains, 50–70%), Surface creep (rolling coarse grains), Suspension (fine dust airborne). Water erosion: Sheet → Rill → Gully.

## 7. Energy Resources (Fossil Fuels vs. Renewable Alternatives)

### 7.1 Environmental Fallout of Fossil Fuels

- **Coal Combustion:** Emits large amounts of  $CO_2$  (greenhouse gas driving global warming),  $SO_2$  and  $NO_x$  (causing Acid Rain,  $pH < 5.6$ ), particulate matter, and toxic fly ash containing heavy metals (mercury, lead).
- **Petroleum / Transportation:** Major source of photochemical smog (ground-level ozone,  $PAN$ ), carbon monoxide ( $CO$ ), and marine oil spills.
- **Nuclear Power:** Zero greenhouse gas emissions during operation, but carries catastrophic risk of core meltdown (Chernobyl 1986, Fukushima 2011) and unsolved challenges of long-term deep geological disposal of high-level radioactive waste (plutonium isotopes with half-lives exceeding 24,000 years).

### 7.2 Renewable Energy Alternatives

- **Solar Energy:** Photovoltaic (PV) cells convert sunlight directly into electricity via silicon semiconductor p-n junctions; solar thermal collectors generate heat/steam.
- **Wind Energy:** Aerodynamic wind turbine blades capture atmospheric kinetic energy; clean, modular, zero operational water use.
- **Biomass & Biogas:** Anaerobic digestion of animal manure and organic waste by methanogenic bacteria produces **Biogas** (55 – 70%  $\text{CH}_4$ , 30 – 45%  $\text{CO}_2$ ), providing clean domestic cooking fuel while yielding nitrogen-rich organic slurry fertilizer.
- **Hydroelectric & Geothermal:** Hydro harness gravitational potential of flowing water; Geothermal taps subterranean hydrothermal steam.

#### Quick Recall:

Fossil fuels drive acid rain ( $\text{SO}_2$ ,  $\text{NO}_x$ ) and global warming ( $\text{CO}_2$ ). Renewables (Solar PV, Wind, Biogas [methane]) provide decentralized, low-carbon power.

## 8. Solved Environmental Calculations

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Solved Problem 2.1: Rooftop Rainwater Harvesting Yield:

**Problem Statement:** A residential campus building has a flat concrete rooftop of dimensions  $15 \text{ m} \times 10 \text{ m}$ . The annual rainfall in the region is  $900 \text{ mm}$ . If the runoff coefficient for the concrete roof is  $C = 0.85$ , calculate:

1. The total volume of rainwater that can be harvested annually in cubic meters ( $\text{m}^3$ ).
2. The total volume in liters.

**Step 1: Compute Catchment Roof Area ( $A$ ):**

$$A = 15 \text{ m} \times 10 \text{ m} = 150 \text{ m}^2$$

**Step 2: Convert Rainfall to Meters ( $R$ ):**

$$R = \frac{900 \text{ mm}}{1000 \text{ mm/m}} = 0.90 \text{ m}$$

**Step 3: Apply the Harvesting Formula ( $V = A \times R \times C$ ):**

$$V = 150 \text{ m}^2 \times 0.90 \text{ m} \times 0.85 = 135 \times 0.85 = 114.75 \text{ m}^3$$

**Step 4: Convert to Liters ( $1 \text{ m}^3 = 1,000 \text{ Liters}$ ):**

$$V_{\text{litres}} = 114.75 \times 1,000 = 114,750 \text{ Liters}$$

$$V = 114.75 \text{ m}^3 = 114,750 \text{ Liters / year}$$

**Impact:** This harvested volume can fulfill the 135 L/capita/day domestic drinking and sanitary needs of a 4-person family for over 212 days!

Solved Problem 2.2: Solar Rooftop PV Generation and Annual Carbon Offset:

**Problem Statement:** An institution installs a 5.0 kW<sub>p</sub> rooftop Solar PV system. In Karnataka, each 1 kW<sub>p</sub> generates an average of 4.5 kWh of electricity per day. If the Indian coal-dominated power grid has an average emission factor of 0.82 kg CO<sub>2</sub>/kWh, calculate:

1. The total electrical energy generated by the system annually.
2. The annual carbon dioxide (CO<sub>2</sub>) emission offset in metric tons.

**Step 1: Annual Electricity Generation:**

$$E_{\text{annual}} = 5.0 \text{ kW}_p \times 4.5 \text{ kWh/kW}_p/\text{day} \times 365 \text{ days} = 22.5 \times 365$$

**Step 2: Annual Carbon Dioxide Offset:**

$$\text{Offset} = 8,212.5 \text{ kWh} \times 0.82 \text{ kg CO}_2/\text{kWh} = \mathbf{6,734.25 \text{ kg CO}_2}$$

**Step 3: Convert to Metric Tons (1 metric ton = 1,000 kg):**

$$\text{Offset}_{\text{tons}} = \frac{6,734.25}{1,000} \approx \mathbf{6.734 \text{ metric tons of CO}_2}$$

$$\mathbf{E_{\text{annual}} = 8,212.5 \text{ kWh}, \quad \text{Annual CO}_2 \text{ Offset} = 6.734 \text{ Metric Tons}}$$

## INTERACTIVE LAB EXPLORER

Water Harvesting & Solar Carbon Offset Studio

Tab 1: Rainwater Harvesting

Tab 2: Solar PV &amp; Carbon Offset

Calculate annual rooftop rainwater harvesting potential using the hydrological runoff equation:

$$V = A \times R \times C$$

Roof Catchment Area ( $A$ ):

150

 $\text{m}^2$ 

Annual Rainfall ( $R$ ):

900

 $\text{mm/yr}$  (0.90 m)

Roof Surface / Runoff Coeff ( $C$ ):

Reinforced Concrete Roof ( $C = 0.85$  - Solved Example)

## HARVESTED WATER YIELD

**114,750** Litres / year

Volume:  $114.75 \text{ m}^3$  ( $150 \times 0.90 \times 0.85$ )

**Domestic Water Security Index:**

Supplies a 4-person household (at 135 L/capita/day = 540 L/day) for **212.5 days** of clean domestic use!

## 9. Quick Revision Sheet

| Resource Topic | Degradation Mechanism                                                      | Mitigation & Conservation Strategy                                                  |
|----------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Forests        | Deforestation, mining, mega-dams submergence.                              | Afforestation, Chipko Movement, Appiko Movement.                                    |
| Water          | Over-pumping, water table collapse, river disputes (Cauvery).              | Rainwater Harvesting ( $V = A \times R \times C$ ), recharge pits, drip irrigation. |
| Minerals       | Acid Mine Drainage ( $H_2SO_4$ generation), tailing dam failures.          | Mine reclamation, lime neutralization of AMD, mineral recycling.                    |
| Food & Soil    | Eutrophication, Biomagnification (DDT), Salinization, Erosion (Saltation). | Organic farming, contour terracing, shelterbelts, mulching.                         |
| Energy         | Fossil fuel acid rain ( $SO_2$ , $NO_x$ ), nuclear radioactive wastes.     | Solar PV, Wind, Biogas (methanogenic anaerobic digestion).                          |

## 10. Authentic Exam Archive & Model Solutions

The following authentic questions have been transcribed directly from official department examination papers in notes/evs/practice/evs-cie1-qp-2024.docx (Ramaiah Continuous Internal Evaluation - I, Course Code HS 510: Environmental Studies):

### Part A: 1-Mark Objective Questions (CIE-1 2024-2025)

 Practice Problem:

CIE-1 Q4 [1M] These are some of the adverse effects of mining:

Solution:

**Official Options:** (a) Water pollution · (b) Air pollution · (c) Noise pollution · (d) All of the above

**Correct Answer:** (d) All of the above.

**Academic Explanation:** Mining activities induce multiple cross-media ecological damage:

- *Water Pollution:* Acid Mine Drainage (oxidation of iron pyrite  $FeS_2$ ), leaching of toxic heavy metals ( $Pb$ ,  $Cd$ ,  $As$ ), and siltation of downstream rivers from tailings pond breaches.
- *Air Pollution:* Fugitive particulate matter emissions ( $PM_{10}$  and  $PM_{2.5}$ ) generated by blasting, haul roads, and mineral processing crushers.
- *Noise Pollution:* High-decibel emissions and ground vibrations caused by explosive blasting, heavy earth-moving equipment, and slurry transport pumps.

 Practice Problem:

CIE-1 Q5 [1M] Saltation, surface creep, and suspension are the types of soil erosion by:

Solution:

**Official Options:** (a) Wind · (b) Water · (c) Mining · (d) None of the above

**Correct Answer:** (a) Wind.

**Physical Transport Mechanics:**

- **Saltation (50–70% of wind transport):** Medium sand particles (0.1 – 0.5 mm) bounce in short hops along the ground.
- **Surface Creep (5–25% of transport):** Coarser soil grains (0.5 – 2.0 mm) are pushed or rolled along the surface by colliding saltating particles.
- **Suspension (3–40% of transport):** Very fine silt and clay particles ( $< 0.1$  mm) are lifted into turbulent upper atmospheric air currents and transported over thousands of kilometers.

 Practice Problem:

CIE-1 Q7 [1M] Extensive planting of trees to increase forest cover is called:

Solution:

**Official Options:** (a) Afforestation · (b) Agro forestry · (c) Deforestation · (d) Social forestry

**Correct Answer:** (a) Afforestation.

**Explanation:** Afforestation is the establishment of a forest or stand of trees in an area where there was no recent previous tree cover, serving as a carbon sink and halting desertification.

 Practice Problem:

CIE-1 Q8 [1M] Sustainability requires:

Solution:

**Official Options:** (a) Conservation of resources · (b) Minimizing depletion of non-renewable resources · (c) Using sustainable practices for managing renewable resources · (d) All

**Correct Answer:** (d) All.

**Explanation:** Ecological sustainability requires balancing consumption with regeneration rates so present societal needs are met without compromising future generations.

 Practice Problem:

CIE-1 Q9 [1M] Which of the following is the most environmentally friendly agriculture practice?

Solution:

**Official Options:** (a) Using chemical fertilizers · (b) Using insecticides · (c) Organic farming · (d) None of the above

**Correct Answer:** (c) Organic farming.

**Explanation:** Organic farming eliminates synthetic agrochemicals, relying on biofertilizers (compost, vermicompost, *Rhizobium*), crop rotation, and biological pest control to preserve soil microbial biodiversity and prevent groundwater nitrate toxicity.

 Practice Problem:

CIE-1 Q10 [1M] A mineral is defined as a:

Solution:

**Official Options:** (a) Organic matter · (b) Naturally occurring inorganic substance · (c) Synthetic compound · (d) None

**Correct Answer:** (b) Naturally occurring inorganic substance.

**Explanation:** A mineral is a naturally occurring, homogeneous, inorganic solid having a definite chemical composition and an ordered internal crystalline structure.

Part B: 5-Mark Subjective Questions (CIE-1 2024-2025) Practice Problem:

CIE-1 Q1.b [5M] What are the environmental effects of using fossil fuels?

Solution:

**Official Model Solution:** The combustion of fossil fuels (coal, petroleum, natural gas) generates severe multidimensional impacts:

1. **Anthropogenic Greenhouse Effect & Global Warming:** Burning carbon yields massive  $CO_2$  emissions (rising from pre-industrial 280 ppm to over 420 ppm), driving global temperature rise, polar ice melt, and sea-level rise.
2. **Acid Deposition (Acid Rain):** High sulfur and nitrogen content in coal/petroleum combusts into  $SO_2$  and  $NO_x$ , which oxidize into sulfuric acid ( $H_2SO_4$ ) and nitric acid ( $HNO_3$ ), dropping precipitation  $pH < 5.6$  and corroding monuments and acidifying freshwater fisheries.
3. **Photochemical Smog & Respiratory Distress:** Vehicle exhausts release volatile organic compounds (VOCs) and  $NO_x$ , reacting with sunlight to form ground-level ozone ( $O_3$ ) and photochemical smog, triggering asthma and chronic bronchitis.
4. **Thermal Pollution & Ash Disposal:** Coal-fired thermal power stations produce millions of tons of fly ash containing toxic heavy metals ( $Pb$ ,  $As$ ,  $Hg$ ), contaminating groundwater aquifers and topsoils.

 **Practice Problem:**

CIE-1 Q2.b [5M] Explain the effects of overuse of fresh water and problems related to water resources.

**Solution:**

**Official Model Solution:** Over-exploitation of freshwater leads to critical hydrological crises:

1. **Water Table Depletion:** Intensive tube-well extraction for green-revolution agriculture exceeds natural monsoon recharge, causing regional aquifer water tables to fall by meters annually.
2. **Saltwater Intrusion:** Excessive drawdown in coastal areas reverses hydraulic gradients, pulling dense saline seawater into freshwater municipal aquifers and ruining drinking water supplies.
3. **Land Subsidence:** Over-extraction drains interstitial pore water from deep silt/clay strata, causing the overburden ground to compact and sink irreversibly (e.g., Mexico City, Jakarta, Venice).
4. **Waterlogging and Soil Salinization:** In canal-irrigated zones, excessive irrigation without adequate drainage raises the water table into the root zone, causing capillary action to evaporate water and leave a toxic crust of alkaline salts on arable soils.
5. **Riparian & Interstate Water Conflicts:** Upstream impoundments restrict downstream flows during dry seasons, causing severe socio-economic disputes (e.g. Cauvery River dispute between Karnataka and Tamil Nadu).

| <i>Dimension</i>         | <i>Conventional / Non-Renewable</i>                                                 | <i>Non-Conventional / Renewable</i>                                |
|--------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Sources</b>           | Coal, Crude Oil, Natural Gas, Nuclear Fission ( $U^{235}$ ).                        | Solar PV, Wind Turbines, Small Hydro, Biomass, Geothermal.         |
| <b>Depletion Profile</b> | Finite geological deposits; exhausting within decades to centuries.                 | Naturally replenished continuously via solar and planetary cycles. |
| <b>Carbon Footprint</b>  | High greenhouse gas and particulate emissions (0.82 – 1.0 kg CO <sub>2</sub> /kWh). | Zero or near-zero operational greenhouse gas emissions.            |

| Dimension          | Conventional / Non-Renewable                              | Non-Conventional / Renewable                                     |
|--------------------|-----------------------------------------------------------|------------------------------------------------------------------|
| Ecological Hazards | Acid rain, oil spills, radioactive nuclear waste storage. | Intermittent output, land requirement, bird strike risks (wind). |

### Verified Numerical & Conceptual Practice Problems

#### PRACTICE 1 · ROOFTOP RAINWATER HARVESTING YIELD

Calculate the annual harvested rainwater yield for a concrete rooftop of area  $150 \text{ m}^2$  located in a region with  $900 \text{ mm}$  annual rainfall and runoff coefficient  $0.85$ .

#### Practice Problem:

Show Step-by-Step Solution

#### Solution:

Applying the hydrological harvesting equation:  $V = A \times R \times C$

- Catchment Area:  $A = 150 \text{ m}^2$
- Rainfall:  $R = 900 \text{ mm} = 0.90 \text{ m}$
- Runoff Coefficient:  $C = 0.85$  (concrete roof)

$$V = 150 \text{ m}^2 \times 0.90 \text{ m} \times 0.85 = 135 \times 0.85 = \mathbf{114.75 \text{ m}^3}$$

$$V_{\text{litres}} = 114.75 \times 1,000 = \mathbf{114,750 \text{ Litres}}$$

Verification: Verified by automated unit test in  
audit/verify/evs/verify\_evs\_u2.py.

PRACTICE 2 · SOLAR PV GENERATION & CO<sub>2</sub> OFFSET

A 5.0 kW<sub>p</sub> solar PV rooftop system operates with an average yield of 4.5 kWh/kW<sub>p</sub>/day. If the grid emission factor is 0.82 kg CO<sub>2</sub>/kWh, calculate annual energy generated and metric tons of CO<sub>2</sub> offset.

 Practice Problem:

Show Step-by-Step Solution

Solution:

## 1. Annual Electricity Generation:

$$E_{\text{annual}} = 5.0 \text{ kW}_p \times 4.5 \text{ kWh/kW}_p/\text{day} \times 365 = \mathbf{8,212.5}$$

## 2. Annual Carbon Dioxide Offset:

$$\text{CO}_2 \text{ Offset} = 8,212.5 \text{ kWh} \times 0.82 \text{ kg CO}_2/\text{kWh} = \mathbf{6,734.25}$$

## 3. Conversion to Metric Tons:

$$\text{Offset}_{\text{tons}} = \frac{6,734.25}{1,000} = \mathbf{6.734} \text{ Metric Tons}$$

Verification: Verified by automated unit test in  
audit/verify/evs/verify\_evs\_u2.py.

**PRACTICE 3 · ACID MINE DRAINAGE (AMD) REACTION**

*Write the stoichiometric chemical reaction for Acid Mine Drainage (AMD) formation and explain why it severely pollutes aquatic ecosystems.*

 Practice Problem:

Show Step-by-Step Solution

Solution:

When sub-surface mining exposes iron pyrite ( $FeS_2$ ) to atmospheric oxygen and groundwater, iron-oxidizing bacteria (*Acidithiobacillus ferrooxidans*) catalyze the reaction:



**Ecological Impact:** Generates concentrated sulfuric acid ( $H_2SO_4$ ), driving stream *pH* down to 2 – 3, dissolving highly toxic heavy metals (aluminum, lead, cadmium), and suffocating aquatic gills with insoluble orange-brown ferric hydroxide precipitations ("yellow boy").

**PRACTICE 4 · WATERLOGGING VS SALINIZATION MECHANISMS**

*Differentiate between waterlogging and soil salinization in agricultural land management.*

 Practice Problem:

Show Step-by-Step Solution

Solution:

- **Waterlogging:** Physical condition where excessive irrigation water saturates soil pore spaces, pushing the water table into the plant root zone, displacing oxygen, and causing anaerobic root asphyxiation.
- **Soil Salinization:** Chemical degradation where high water tables evaporate in arid/semi-arid climates, drawing dissolved minerals upward via capillary action and depositing a sterile white crust of alkaline sodium, calcium, and magnesium salts on topsoil.

| Mode of Erosion         | Particle Diameter Range   | Percentage of Total Wind Transport |
|-------------------------|---------------------------|------------------------------------|
| Saltation (Bouncing)    | 0.1 mm to 0.5 mm          | 50% to 70% (Dominant mechanism)    |
| Surface Creep (Rolling) | 0.5 mm to 2.0 mm          | 5% to 25%                          |
| Suspension (Airborne)   | < 0.1 mm (Fine silt/clay) | 3% to 40%                          |

#### PRACTICE 6 · ECOLOGICAL CONFLICTS OF MEGA DAMS

*Analyze the key environmental and social trade-offs of large river valley projects with reference to Sardar Sarovar and Tehri Dams.*

##### Practice Problem:

Show Step-by-Step Solution

##### Solution:

While multipurpose mega dams generate hydroelectricity and irrigation water, they incur severe ecological and socio-economic trade-offs:

- **Forest & Biodiversity Submergence:** Inundation of vast tracts of pristine riverine forests, destroying wildlife corridors and endemic habitats.
- **Tribal Displacement & Resettlement Trauma:** Uprooting indigenous tribal communities from ancestral lands without adequate rehabilitation (the core focus of Medha Patkar's *Narmada Bachao Andolan*).
- **Reservoir-Induced Seismicity (RIS):** Massive concentrated water masses in seismic zones (e.g. Tehri Dam in the Central Himalayan seismic gap) create pore pressure lubrication along active fault lines.
- **Silt Trap & Downstream Delta Erosion:** Trapping fertile nutrient silts in reservoirs deprives downstream estuaries, starving agricultural deltas and triggering coastal erosion.

## 11. Prescribed References

### Course Reference Literature:

1. Erach Bharucha, *Textbook of Environmental Studies for Undergraduate Courses*, Universities Press / UGC India. (Chapter 2: Natural Resources).
2. Benny Joseph, *Environmental Studies*, Tata McGraw-Hill Education.
3. R. Rajagopalan, *Environmental Studies: From Crisis to Cure*, Oxford University Press.

*AI-assisted study notes written from the official courseware. Verify against your faculty's presentation slides before the exam.*